EWB CLOUD LABS: STUDENT PROJECT WORKBOOK



AWS VPC Peering - Build / Test / Explain

Name: Jayasurya

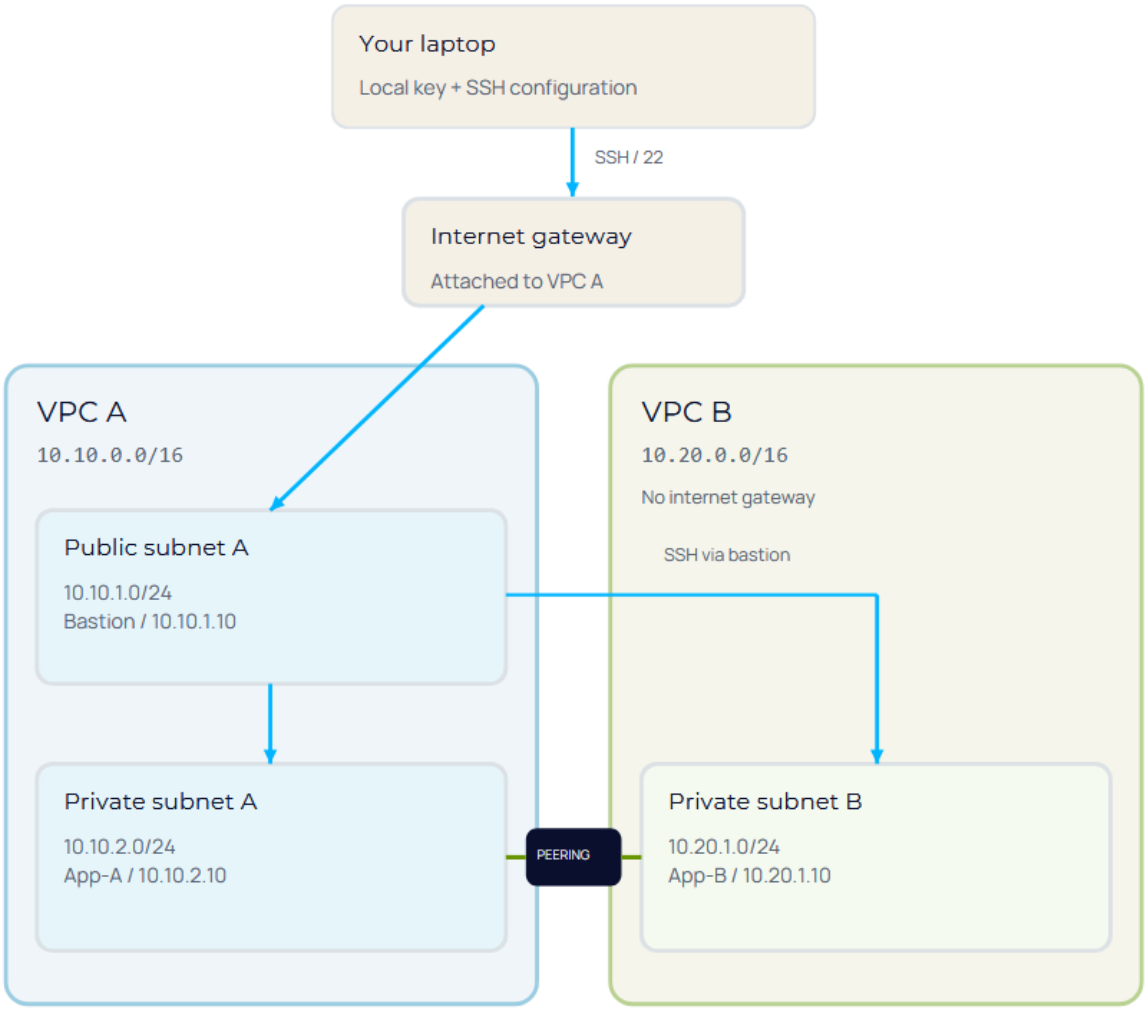
Batch: 01

Project Title: Connect two VPCs. Prove private connectivity.

Region: Asia Pacific (Mumbai) / ap-south-1

Date: 29 October 2026

1. Resource Inventory & Architecture



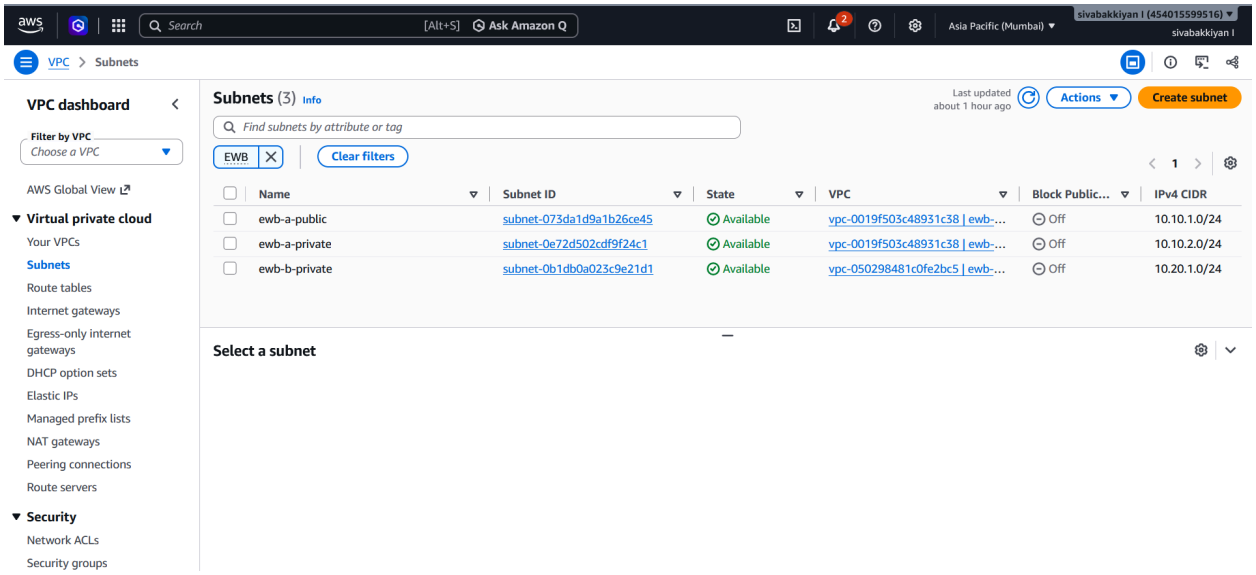
Resource Name	Type	CIDR / IP Address	Actual Resource ID
ewb-vpc-a	VPC A	10.10.0.0/16	vpc-0019f503c48931c38
ewb-vpc-b	VPC B	10.20.0.0/16	vpc-050298481c0fe2bc5
ewb-a-public	Subnet (VPC A)	10.10.1.0/24	subnet-073da1d9a1b26ce45
ewb-a-private	Subnet (VPC A)	10.10.2.0/24	subnet-0e72d502cdf9f24c1

ewb-b-private	Subnet (VPC B)	10.20.1.0/24	subnet-0b1db0a023c9e21d1
ewb-a-b-peer	Peering Connection	N/A	pcx-0ca104a427c21005b
ewb-bastion	EC2 Instance	10.10.1.10 (Private)	i-08ea40a3c3cd90f1c
ewb-app-a	EC2 Instance	10.10.2.10 (Private)	i-0ca7e54b41e54b870
ewb-app-b	EC2 Instance	10.20.1.10 (Private)	i-079e004bddf4d2865

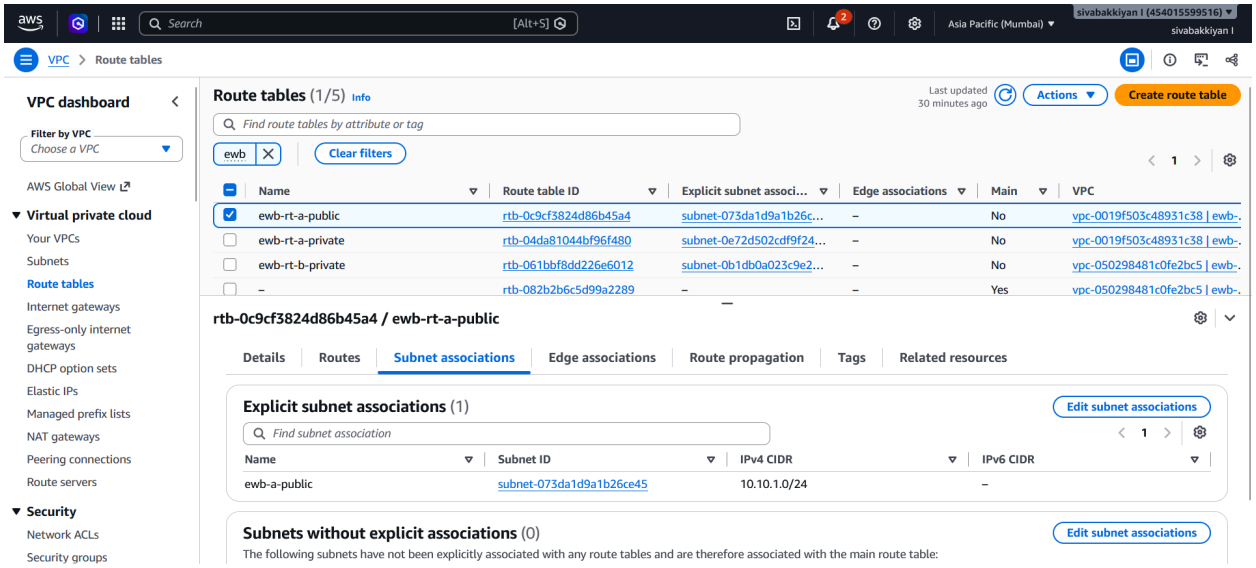
2. Configuration Screenshots

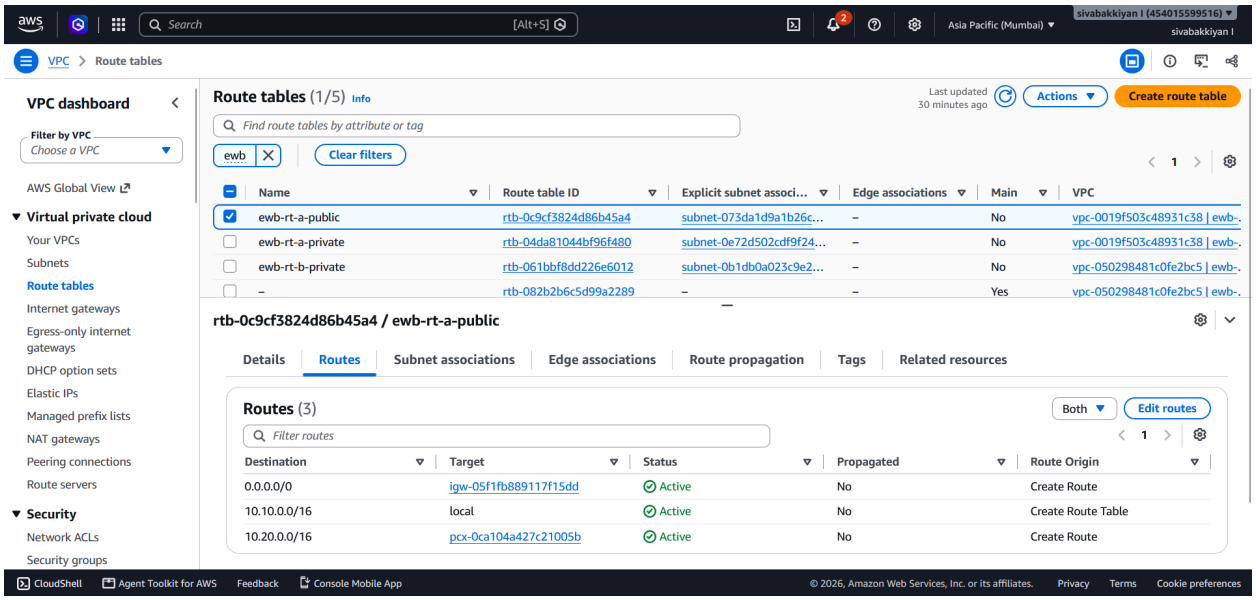
A. Subnets & Route Tables Configuration:

1) SUBNET SCREENSHOTS

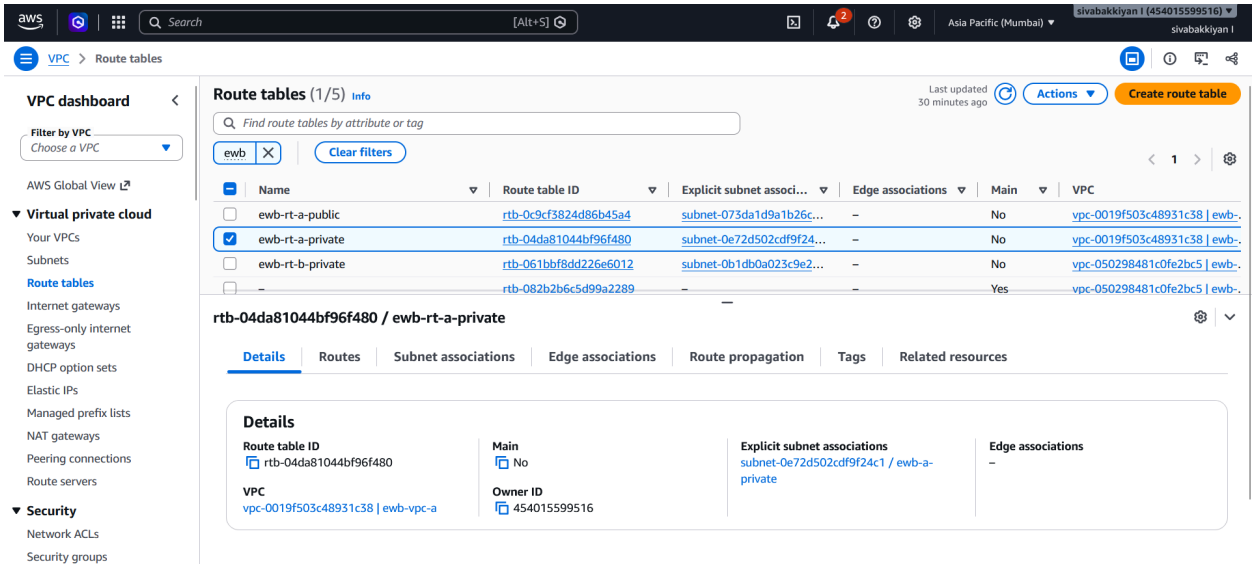


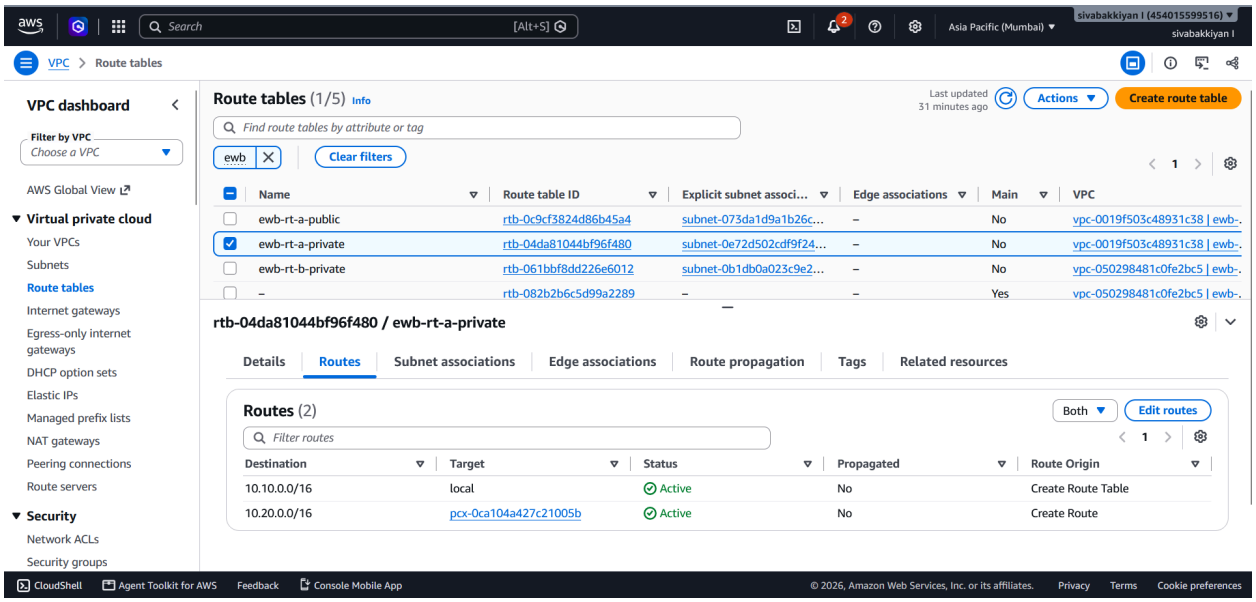
2) PUBLIC ROUTE TABLE SCREENSHOTS

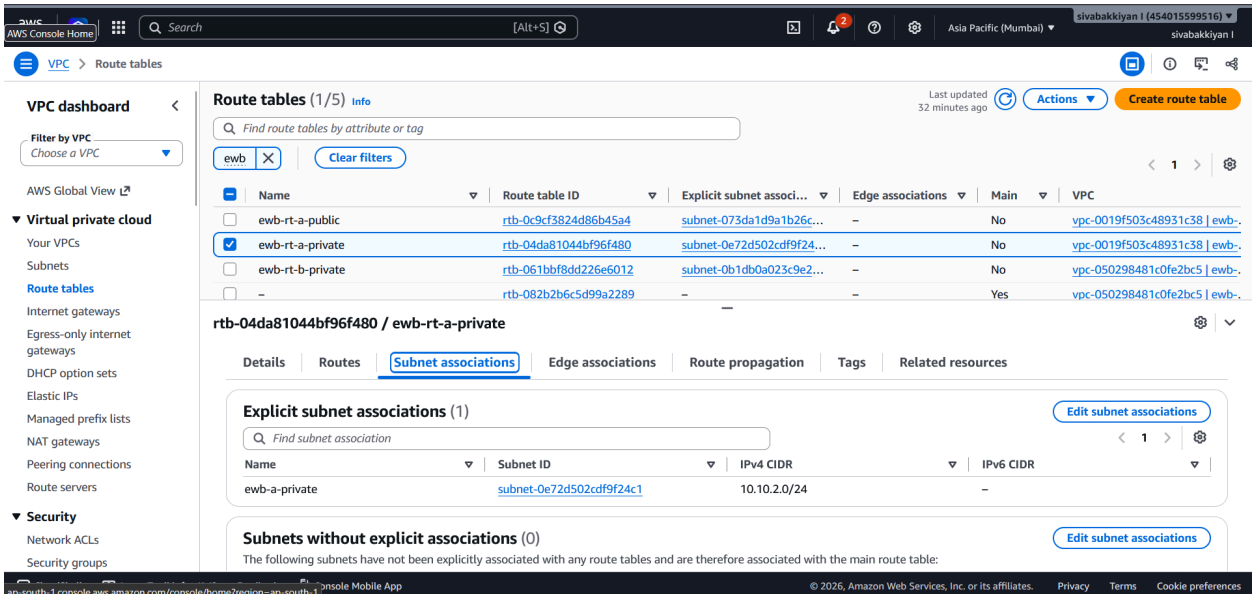


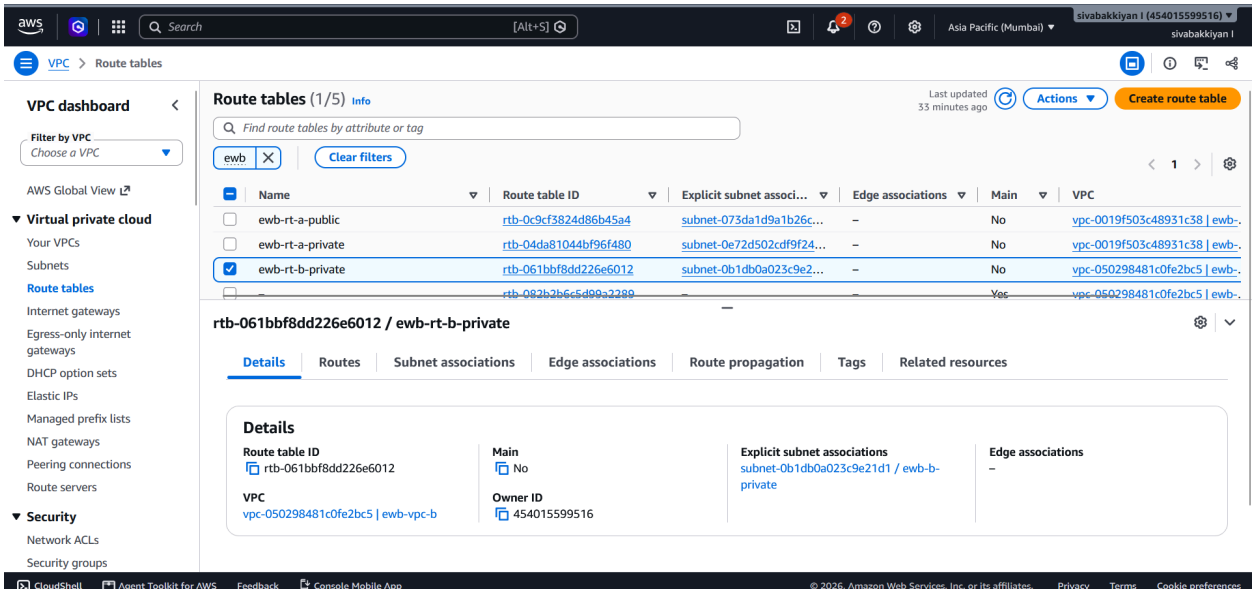


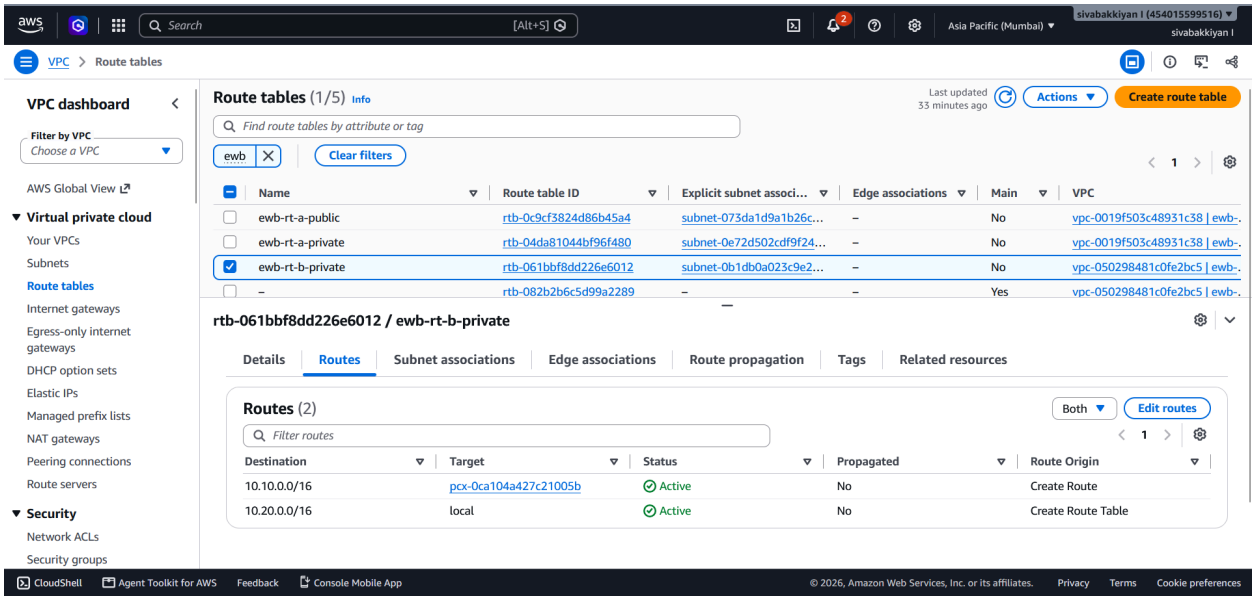
3) PRIVATE ROUTE TABLES SCREENSHOTS

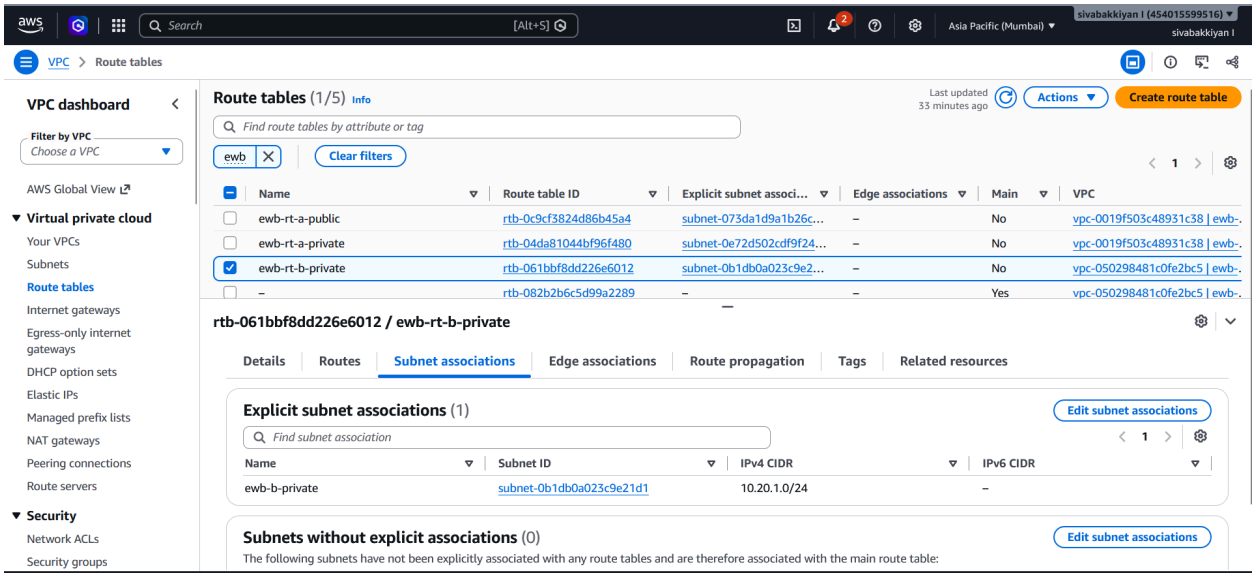

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Route tables
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DHCP option sets
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Peering connections
Route servers
Security
Network ACLs
Security groups
Route tables (1/5) Info
Last updated 30 minutes ago
Actions
Create route table
Find route tables by attribute or tag
ewb
Clear filters
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
ewb-rt-a-public
rtb-0c9cf3824d86b45a4
subnet-073da1d9a1b26c...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-a-private
rtb-04da81044bf96f480
subnet-0e72d502cdf9f24...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-b-private
rtb-061bbf8dd226e6012
subnet-0b1db0a023c9e2...
–
No
vpc-050298481c0fe2bc5 | ewb-...
–
rtb-082b2b6c5d99a2289
–
–
Yes
vpc-050298481c0fe2bc5 | ewb-...
rtb-04da81044bf96f480 / ewb-rt-a-private
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Details
Route table ID
rtb-04da81044bf96f480
Main
No
Explicit subnet associations
subnet-0e72d502cdf9f24c1 / ewb-a-private
Edge associations
–
VPC
vpc-0019f503c48931c38 | ewb-vpc-a
Owner ID
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DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
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Network ACLs
Security groups
Route tables (1/5) Info
Last updated 31 minutes ago
Actions
Create route table
Find route tables by attribute or tag
ewb
Clear filters
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
ewb-rt-a-public
rtb-0c9cf3824d86b45a4
subnet-073da1d9a1b26c...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-a-private
rtb-04da81044bf96f480
subnet-0e72d502cdf9f24...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-b-private
rtb-061bbf8dd226e6012
subnet-0b1db0a023c9e2...
–
No
vpc-050298481c0fe2bc5 | ewb-...
–
rtb-082b2b6c5d99a2289
–
–
Yes
vpc-050298481c0fe2bc5 | ewb-...
rtb-04da81044bf96f480 / ewb-rt-a-private
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
10.10.0.0/16
local
Active
No
Create Route Table
10.20.0.0/16
pcx-0ca104a427c21005b
Active
No
Create Route
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DHCP option sets
Elastic IPs
Managed prefix lists
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Peering connections
Route servers
Security
Network ACLs
Security groups
Route tables (1/5) Info
Last updated 32 minutes ago
Actions
Create route table
Find route tables by attribute or tag
ewb
Clear filters
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
ewb-rt-a-public
rtb-0c9cf3824d86b45a4
subnet-073da1d9a1b26c...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-a-private
rtb-04da81044bf96f480
subnet-0e72d502cdf9f24...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-b-private
rtb-061bbf8dd226e6012
subnet-0b1db0a023c9e2...
–
No
vpc-050298481c0fe2bc5 | ewb-...
–
rtb-082b2b6c5d99a2289
–
–
Yes
vpc-050298481c0fe2bc5 | ewb-...
rtb-04da81044bf96f480 / ewb-rt-a-private
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Explicit subnet associations (1)
Edit subnet associations
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
ewb-a-private
subnet-0e72d502cdf9f24c1
10.10.2.0/24
–
Subnets without explicit associations (0)
Edit subnet associations
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:
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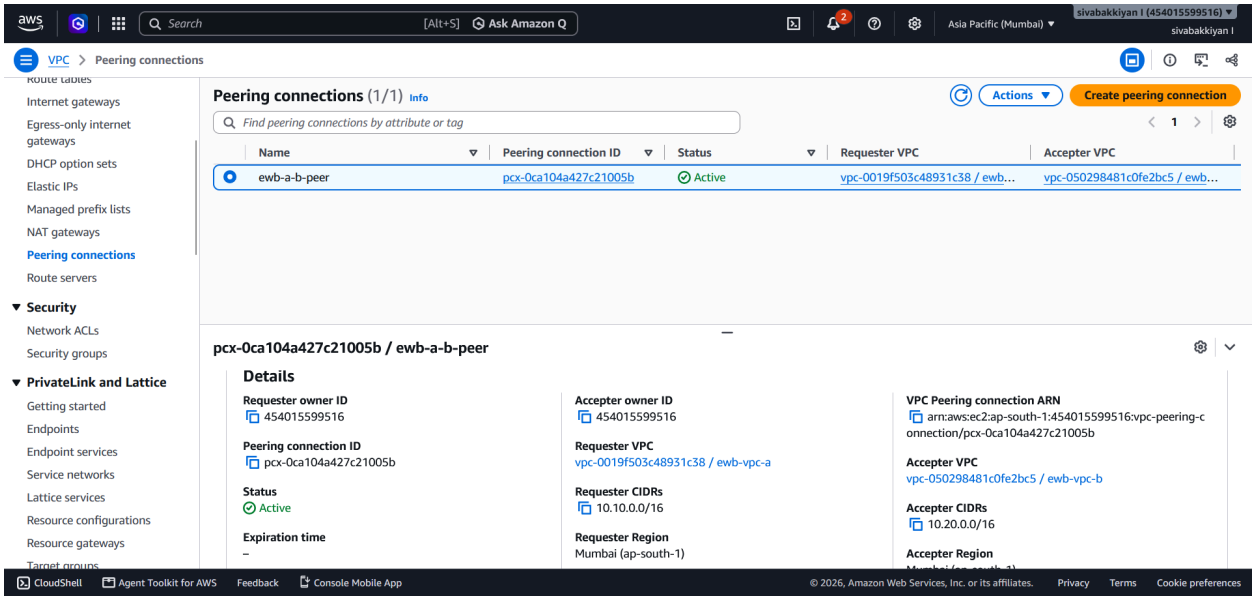

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Security
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Route tables (1/5) Info
Last updated 33 minutes ago
Actions
Create route table
Find route tables by attribute or tag
ewb
Clear filters
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
ewb-rt-a-public
rtb-0c9cf3824d86b45a4
subnet-073da1d9a1b26c...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-a-private
rtb-04da81044bf96f480
subnet-0e72d502cdf9f24...
–
No
vpc-0019f503c48931c38 | ewb-...
ewb-rt-b-private
rtb-061bbf8dd226e6012
subnet-0b1db0a023c9e2...
–
No
vpc-050298481c0fe2bc5 | ewb-...
rtb-061bbf8dd226e6012 / ewb-rt-b-private
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Details
Route table ID
rtb-061bbf8dd226e6012
Main
No
Explicit subnet associations
subnet-0b1db0a023c9e21d1 / ewb-b-private
Edge associations
–
VPC
vpc-050298481c0fe2bc5 | ewb-vpc-b
Owner ID
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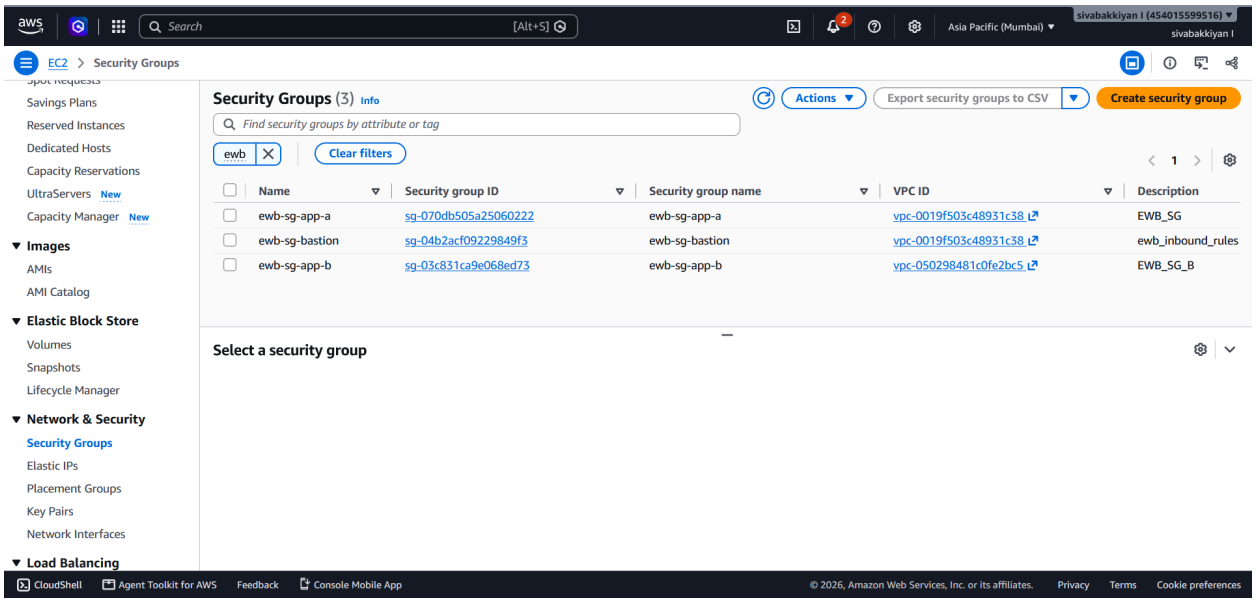
B. Active Peering Connection:

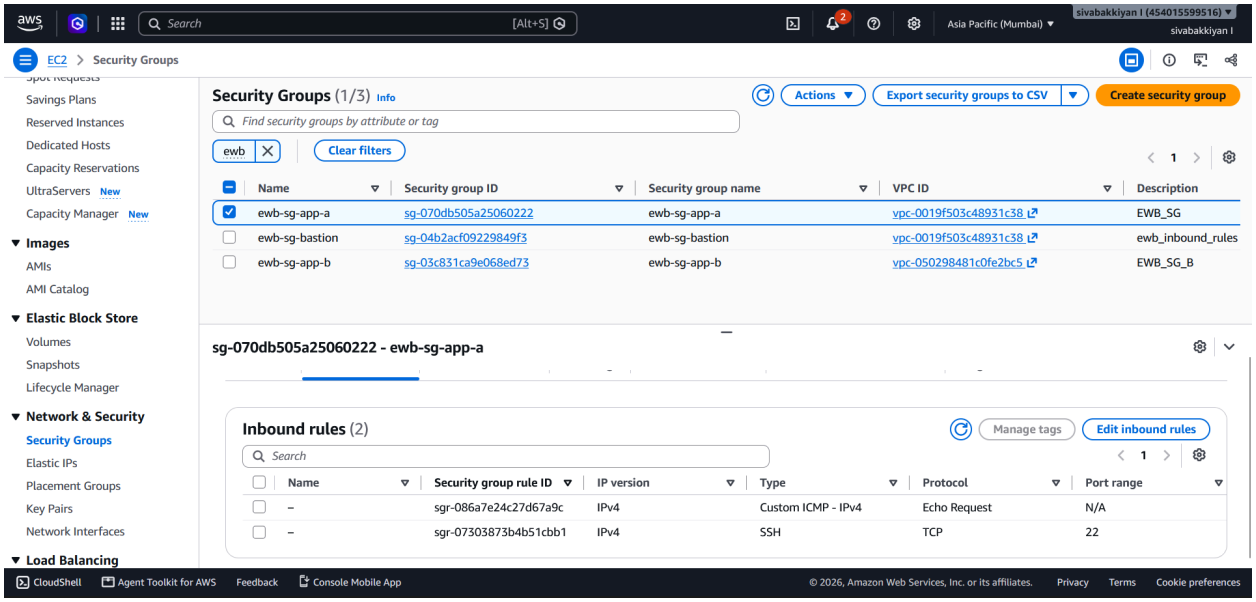
1) ACTIVE PEERING SCREENSHOTS

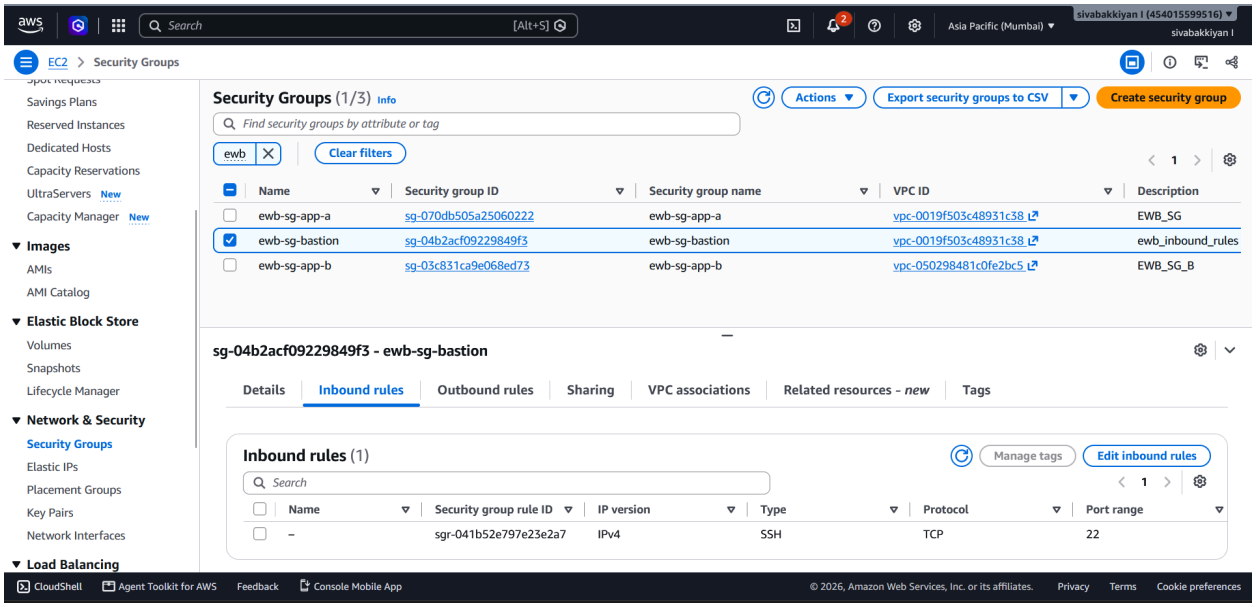


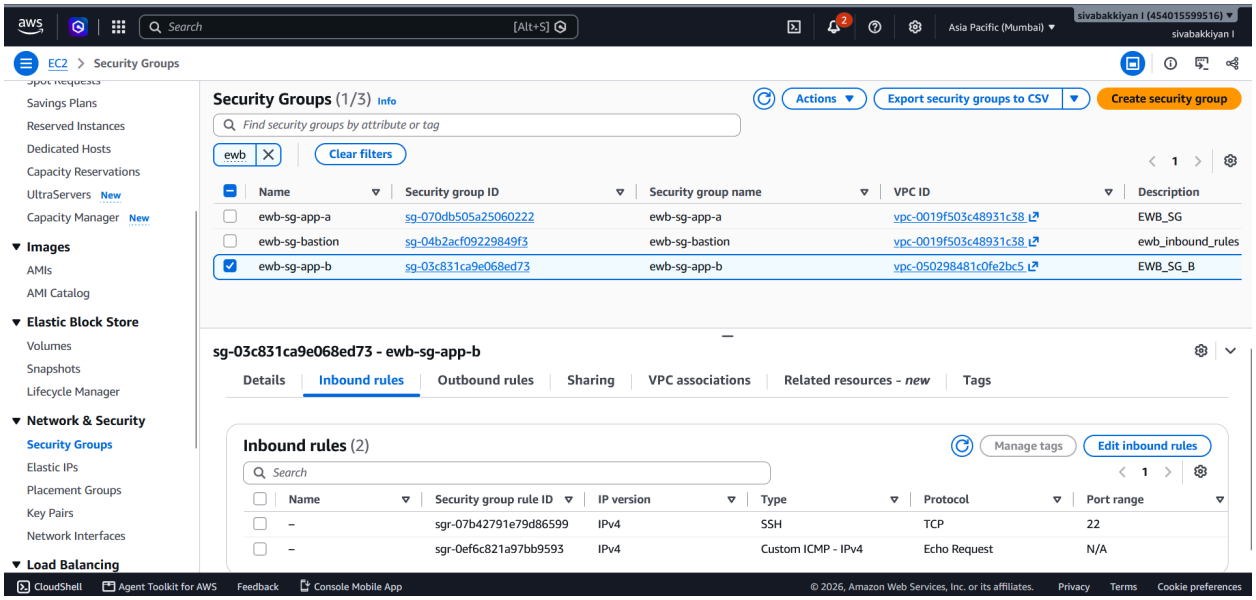
C. Security Group Rules:

1) APP-A and APP-B INBOUND RULES SCREENSHOTS

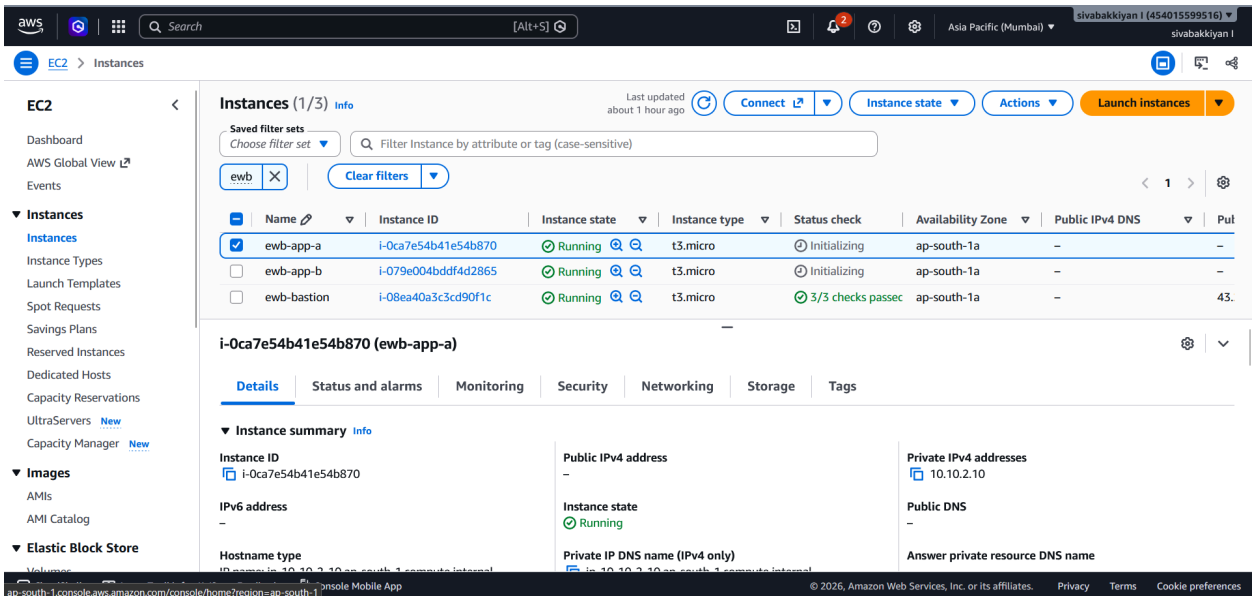


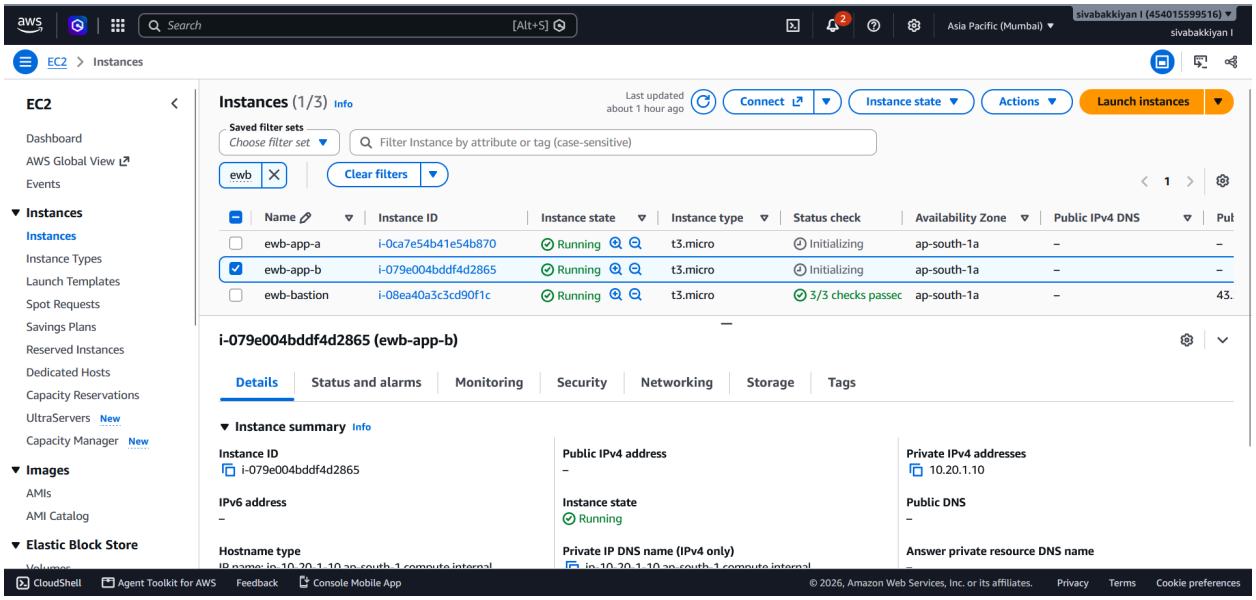

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Security group name
VPC ID
Description
ewb-sg-app-a
sg-070db505a25060222
ewb-sg-app-a
vpc-0019f503c48931c38
EWB_SG
ewb-sg-bastion
sg-04b2acf09229849f3
ewb-sg-bastion
vpc-0019f503c48931c38
ewb_inbound_rules
ewb-sg-app-b
sg-03c831ca9e068ed73
ewb-sg-app-b
vpc-050298481c0fe2bc5
EWB_SG_B
sg-070db505a25060222 - ewb-sg-app-a
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
–
sgr-086a7e24c27d67a9c
IPv4
Custom ICMP - IPv4
Echo Request
N/A
–
sgr-07303873b4b51cbb1
IPv4
SSH
TCP
22
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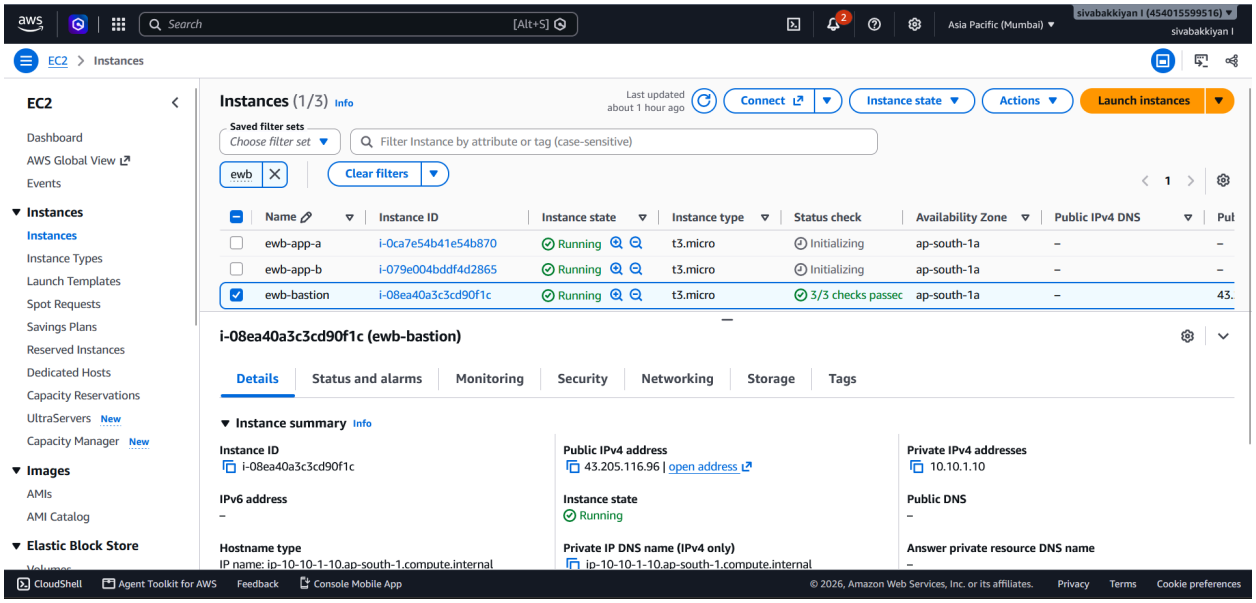

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Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
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Security Groups (1/3) Info
Actions
Export security groups to CSV
Create security group
Find security groups by attribute or tag
ewb
Clear filters
Name
Security group ID
Security group name
VPC ID
Description
ewb-sg-app-a
sg-070db505a25060222
ewb-sg-app-a
vpc-0019f503c48931c38
EWB_SG
ewb-sg-bastion
sg-04b2acf09229849f3
ewb-sg-bastion
vpc-0019f503c48931c38
ewb_inbound_rules
ewb-sg-app-b
sg-03c831ca9e068ed73
ewb-sg-app-b
vpc-050298481c0fe2bc5
EWB_SG_B
sg-04b2acf09229849f3 - ewb-sg-bastion
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
–
sgr-041b52e797e23e2a7
IPv4
SSH
TCP
22
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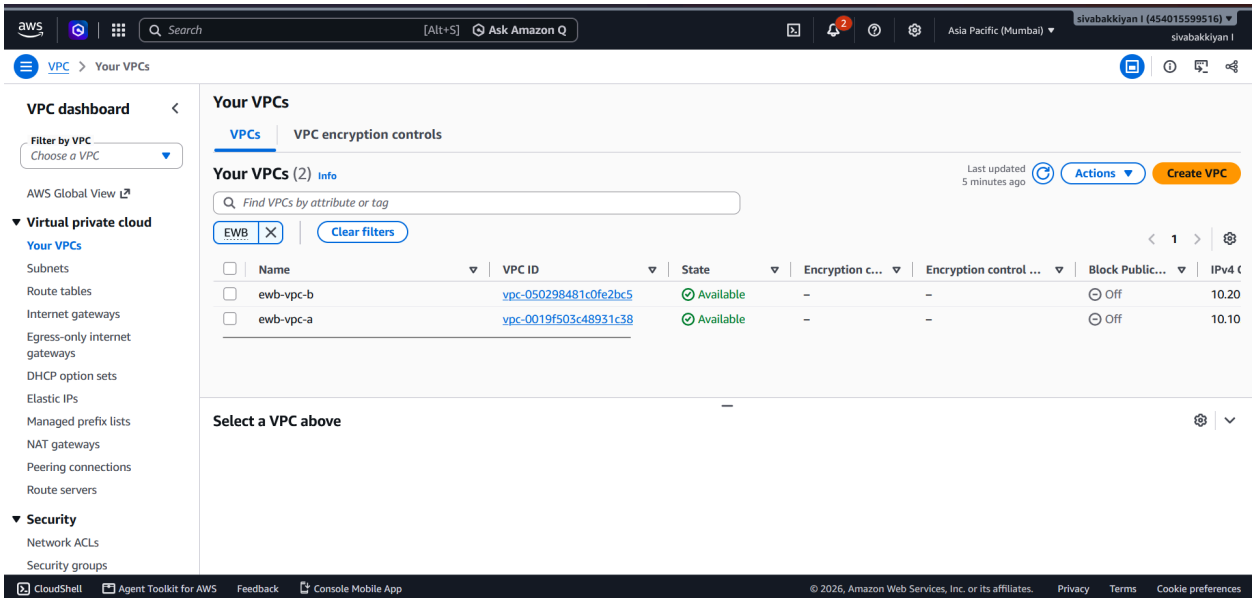
D. EC2 SCREENSHOTS:



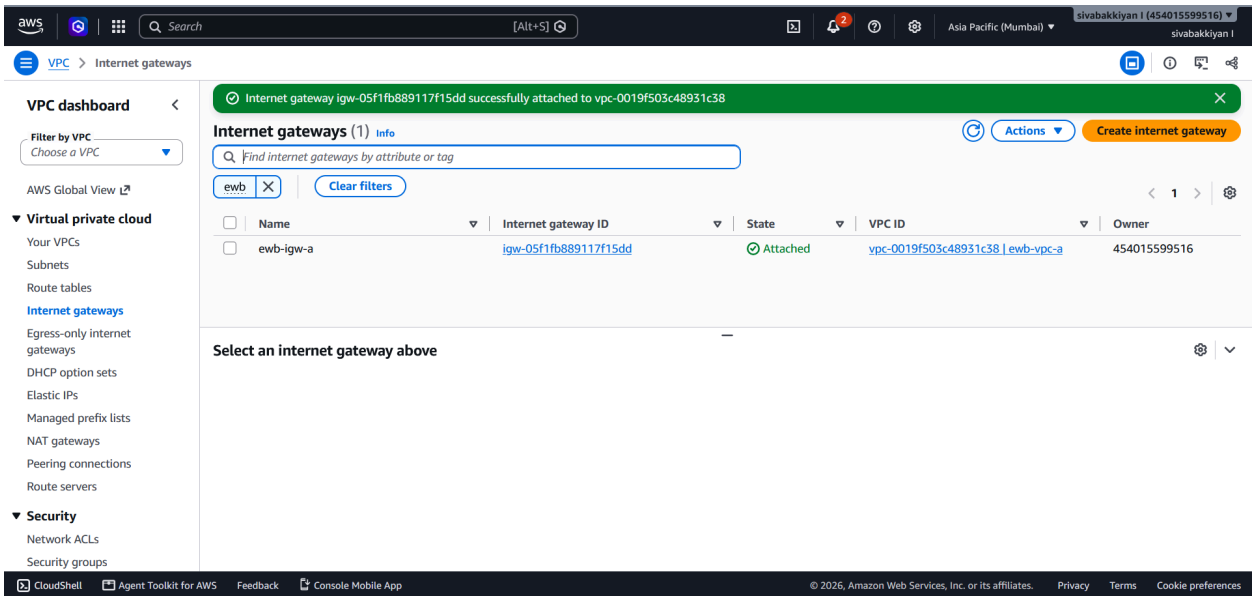




E. VPC SCREENSHOTS:



F. INTERNET GATEWAY SCREENSHOTS:



3. Testing Proofs & Explanations

Test 01: Baseline Failure

1) BASELINE FAILURE PING SCREENSHOTS

```
js171@JayaSurya:/mnt/c/Users/js171/Downloads$ ssh ewb-app-a
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:L1ink6ozLy4uDIn8aNW40+ZR11kJJ4JHIf5lSoegoSM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3086ms

[ec2-user@ip-10-10-2-10 ~]$
```

Explanation: Before setting up the VPC Peering and routing, App-A could not reach App-B. The ping resulted in 100% packet loss because there was no route connecting the two isolated VPC networks.

Test 02: Forward & Reverse Path Ping Success

1) FORWARD PING SUCCESS SCREENSHOT

```
js171@JayaSurya:/mnt/c/Users/js171$ cd Downloads
js171@JayaSurya:/mnt/c/Users/js171/Downloads$ ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 06:35:00 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.474 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.191 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.291 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.173 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3153ms
rtt min/avg/max/mdev = 0.173/0.282/0.474/0.119 ms
[ec2-user@ip-10-10-2-10 ~]$
```

2) REVERSE PING SUCCESS SCREENSHOT

```
js171@JayaSurya:/mnt/c/Users/js171/Downloads$ ssh ewb-app-b
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:FRK7PDjdtWFOF5a8Gn/o51AQScsGha4MSQd8n20H/ds.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.155 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.156 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.211 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.192 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3084ms
rtt min/avg/max/mdev = 0.155/0.178/0.211/0.023 ms
[ec2-user@ip-10-20-1-10 ~]$
```

Routing Explanation:

- **Forward Path:** When App-A (10.10.2.10) pings App-B (10.20.1.10), the traffic goes to VPC A's private route table. The route table checks the destination (10.20.0.0/16) and forwards the traffic to the Active Peering Connection (pcx-...). App-B's Security Group allows this incoming ICMP traffic.
- **Return Path:** For the ping reply to come back, VPC B's private route table directs the return traffic (10.10.0.0/16) back through the same Peering Connection. Without this return route, the ping would fail.

Test 03: Private SSH Access (ProxyJump)

1) SSH TO APP-B FROM LAPTOP SCREENSHOT

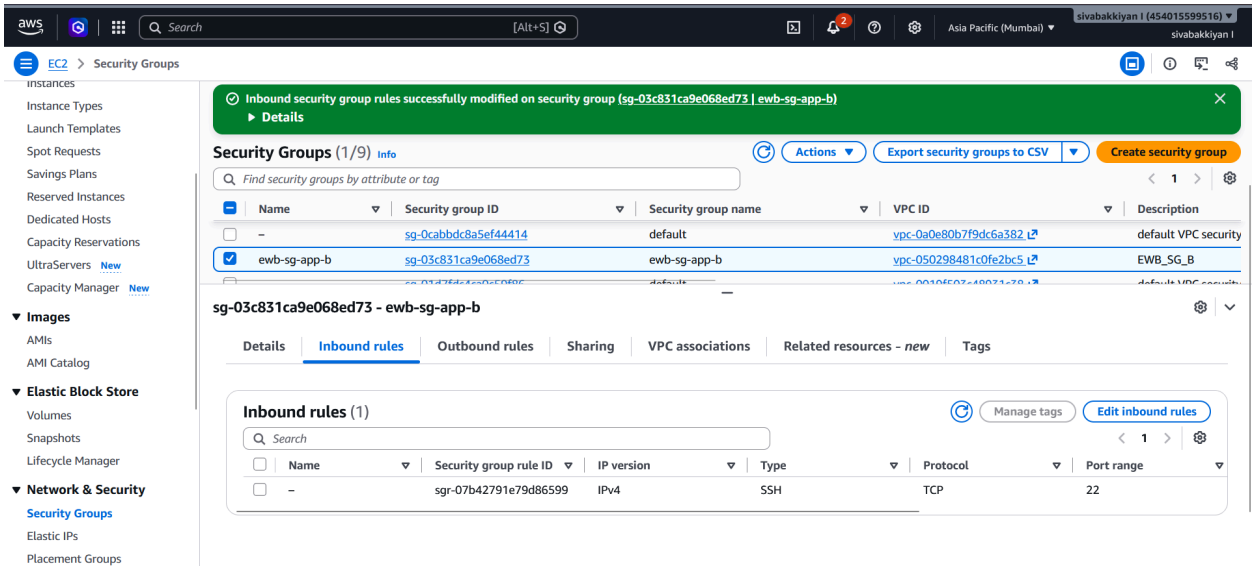
```
js171@JayaSurya:/mnt/c/Users/js171/Downloads$ ssh ewb-app-b
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:FRK7PDjdtWFOF5a8Gn/o51AQScsGha4MSQd8n20H/ds.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.155 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.156 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.211 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.192 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3084ms
rtt min/avg/max/mdev = 0.155/0.178/0.211/0.023 ms
[ec2-user@ip-10-20-1-10 ~]$
```

Explanation: Connected successfully to the private App-B instance from the local laptop using the Bastion host's public IP as a ProxyJump.

Test 04: Controlled ICMP Failure & Recovery

CONTROLLED FAILURE (Ping failing after rule removal) SCREENSHOTS


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default VPC security
ewb-sg-app-b
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```
js171@JayaSurya:/mnt/c/Users/js171/Downloads$ ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 06:47:30 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3084ms

[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3084ms

[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.166 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.244 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.169 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.204 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3114ms
rtt min/avg/max/mdev = 0.166/0.195/0.244/0.031 ms
[ec2-user@ip-10-10-2-10 ~]$
```

Explanation: I temporarily removed the ICMP Echo Request rule from App-B's security group. As a result, the ping from App-A immediately failed. This proves that even with correct routing (VPC Peering), the instance-level firewall (Security Group) must explicitly allow the traffic for communication to happen. Once the rule was added back, the ping recovered successfully.

Declaration: I confirm that all resources (EC2 instances, Peering connections,/IGW, Subnets, and VPCs) created for this lab have been successfully terminated and deleted to prevent unwanted billing.